

Route G2-S/D: Finite Readout and an Energy-Accounted Drive

A conditional scattering prediction, not a particle-mass fit

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Abstract

We broaden the G2-R recoil lines using a declared classical beam and instrumental response. Rate weighting precedes conditioning; optimal sign-classification error measures distinguishability. Separately, we activate a homogeneous periodic modulation of the local quartic. Its energy sidebands carry an explicit pump-work ledger. The drive does not change the free mass tower, and can worsen discrimination. Neither calculation is an absolute finite-encounter probability.

1 Fixed physics and new apparatus assumptions

Retain the G2 fields on $\mathbb{M}^{3,1} \times S^1$, with

$$m_a^2 = M_5^2 + (a + \alpha)^2/R^2, \quad \mu_b^2 = M_D^2 + b^2/R^2, \quad j = n + l - m. \quad (1)$$

The engineering card is $R = 1$, $M_5 = .5$, $\alpha = .25$, $M_D = 5$, $g = .05$, and $n = 1$. The finite internal detector preparation remains $w_l \propto \exp[-l^2/(2(1.2)^2)]$ for $-4 \leq l \leq 4$, normalized to sum to one. All energies and momenta use the same arbitrary mass unit as G2. Nothing here is an experimental calibration.

For resolution, condition on $m = 0$ and detector mass label $r = |j| = 1$. Only $(l, j) = (0, +1), (-2, -1)$ contribute. We assume correct mass labels and COM-frame identification, with equal unit acceptance. Mass confusion, angular/frame uncertainty, backgrounds and efficiency are not included. Incoming momenta are back-to-back with magnitude

$$f_{\sigma_i}(p) = \frac{\mathbf{1}_{p \geq 0} e^{-(p-p_0)^2/(2\sigma_i^2)}}{\sqrt{2\pi}\sigma_i \Phi_N(p_0/\sigma_i)}, \quad p_0 = .2. \quad (2)$$

Here Φ_N is the normal CDF; $f_0 = \delta(p - p_0)$. p_0, σ_i are parent-Gaussian location and scale, not the truncated distribution's actual mean and deviation. For $a = p_0/\sigma_i$ and $\lambda = \varphi(a)/\Phi_N(a)$,

$$\langle p \rangle = p_0 + \sigma_i \lambda, \quad \text{Var}(p) = \sigma_i^2 (1 - a\lambda - \lambda^2). \quad (3)$$

This is an *incoherent classical beam ensemble*, not a coherent four-dimensional wavepacket, duration or luminosity specification. The common external-overlap assumption is retained.

The observed momentum estimator x has response

$$\mathcal{R}_{\sigma_d}(x | p_f) = \frac{e^{-(x-p_f)^2/(2\sigma_d^2)}}{\sqrt{2\pi}\sigma_d}, \quad x \in \mathbb{R}. \quad (4)$$

A negative fitted estimator is allowed; true p_f is nonnegative. Clipping would require a different normalized response. Scan $\sigma_i = 0, .02, .1, .2, .4$ and $\sigma_d = .02, .05, .1, .2, .4$. These are engineering choices, not fitted detector specifications.

2 Forward model and discrimination

Set $E_n = \sqrt{m_n^2 + p^2}$, $E_l = \sqrt{\mu_l^2 + p^2}$, $E_i = E_n + E_l$. For an outgoing total energy E , define

$$p_f(E; m, j) = \frac{\sqrt{[E^2 - (m_m + \mu_j)^2][E^2 - (m_m - \mu_j)^2]}}{2E} \quad (5)$$

only for $E > m_m + \mu_j$; closed channels have zero rate. Undriven $E = E_i$, and the G2 rate is

$$K_{lm}(p) = \frac{g^2 p_f}{16\pi E_n E_l E_i}. \quad (6)$$

The unnormalized readout densities are

$$h_j(x) = w_{j+m-n} \int_0^\infty dp f_{\sigma_i}(p) K_{j+m-n,m}(p) \mathcal{R}_{\sigma_d}(x | p_f), \quad (7)$$

$$Z_j = \int_{\mathbb{R}} h_j(x) dx, \quad Z = Z_+ + Z_-, \quad (8)$$

$$P(j, x | m = 0, r = 1, \text{scattering}) = h_j(x)/Z, \quad \pi_j = Z_j/Z. \quad (9)$$

Rate weighting must precede normalization: averaging normalized fixed- p probabilities first generally gives a different result. Where the denominator is positive, $P(j | x, m, r) = h_j(x)/(h_+(x) + h_-(x))$.

For equal classification costs, deciding the larger $h_j(x)$ minimizes error pointwise. Thus

$$e_* = \frac{1}{Z} \int_{\mathbb{R}} \min(h_+, h_-) dx = \frac{1}{2} \left[1 - \frac{\|h_+ - h_-\|_1}{Z} \right]. \quad (10)$$

The no-momentum baseline is $e_{\text{prior}} = \min(\pi_+, \pi_-)$, not necessarily 1/2. We call $e_* \leq .05$ *resolved* solely as an engineering decision criterion. It is not a discovery significance or universal two-peak criterion. Unknown parameters need nuisance propagation; testing a theory also needs finite event counts.

For $\sigma_i = 0$, the undriven result is an exact two-Gaussian mixture: $u_+ = 1.02175457609$, $u_- = 1.36191317264$. With $\Delta = u_- - u_+ > 0$, its Bayes boundary and error are

$$x_* = \frac{u_+ + u_-}{2} + \frac{\sigma_d^2}{\Delta} \log \frac{\pi_+}{\pi_-}, \quad (11)$$

$$e_* = \pi_+ \left[1 - \Phi_N \left(\frac{x_* - u_+}{\sigma_d} \right) \right] + \pi_- \Phi_N \left(\frac{x_* - u_-}{\sigma_d} \right). \quad (12)$$

This supplies an independent analytic regression. Differentiating $E_i(p) = E_f(p_f)$ gives

$$\frac{dp_f}{dp} = \frac{p(1/E_n + 1/E_l)}{p_f(1/E'_m + 1/E'_j)}, \quad \sigma_{f,j}^2 \simeq \sigma_d^2 + \left(\frac{dp_f}{dp} \right)^2 \sigma_i^2. \quad (13)$$

The width approximation applies only to a narrow beam far from $p = 0$ and thresholds. Our computation uses the nonlinear, rate-weighted integral, not a width fit. Extra Gaussian detector noise cannot improve e_* : convolution contracts the L^1 norm in (10). Changing σ_i changes selected priors and kinematics, so no analogous monotonicity claim is made.

3 Activating a powered local time-dependent background

The new branch replaces only the interaction:

$$S_{\text{int}} = - \int d^4x R d\theta \kappa_5 [1 + \epsilon \cos(\Omega t)] |\Phi|^2 |X|^2, \quad g(t) = g[1 + \epsilon \cos(\Omega t)]. \quad (14)$$

This is a new *prescribed external pump*, not a deduction from the motivating paper or a stabilized vacuum. Choose $\epsilon = .5$, $\Omega = .2$, and inspect $\Omega = .1, .4$ without optimizing a desired line. For $0 \leq \epsilon < 1$ the classical potential is nonnegative at all times. The quadratic action and free mass eigenvalues remain unchanged. This does not establish loop stability or a five-dimensional UV completion.

Spatial and circle homogeneity preserve their momenta and both global charges. The pump rest frame is the incoming COM frame, a physical preferred frame, not Lorentz-invariant energy injection. Expand

$$g(t) = \sum_q g_q e^{-iq\Omega t}, \quad g_0 = g, \quad g_{\pm 1} = g\epsilon/2. \quad (15)$$

At first order,

$$S_{fi}^{(1)} \propto -i(2\pi)^3 \delta^3(\mathbf{P}_f - \mathbf{P}_i) \delta_{n+l, m+j} \sum_{q=0, \pm 1} g_q 2\pi \delta(E_f - E_i - q\Omega), \quad (16)$$

$$E_{lmq} = E_n + E_l + q\Omega, \quad K_{lmq}(p) = \boxed{\frac{|g_q|^2 p_f(E_{lmq}; m, j)}{16\pi E_n E_l E_{lmq}}}. \quad (17)$$

Our $q > 0$ means energy absorbed from the pump. The rate follows from initial normalization $(4E_n E_l)^{-1}$ and integrated two-body phase space $p_f/(4\pi E_{lmq})$. Do not replace the initial energies by outgoing energies. For $q = 0$ it is exactly (6). Reference [2] supplies the established sideband framework; the specific relativistic rate is derived here using [1]'s state/phase-space convention.

Only $q = 0, \pm 1$ occur at this Born order; more vertices can generate higher sidebands. This is a long-time weak-scattering rate, not a unitarized all-orders Floquet solution. A finite rectangular time window instead gives the energy amplitude

$$\sum_q g_q F_T(E_f - E_i - q\Omega), \quad F_T(\omega) = \frac{2 \sin(\omega T/2)}{\omega}, \quad F_T(0) = T. \quad (18)$$

Finite- T cross terms cannot be discarded in general. The rate limit uses $|F_T(\omega)|^2/T \rightarrow 2\pi\delta(\omega)$ distributionally for distinct fixed $\Omega > 0$. The $\Omega \rightarrow 0$ and $T \rightarrow \infty$ limits do not commute: at exactly zero frequency the amplitude is $g(1 + \epsilon)$, not three incoherent terms. Coherent beams or finite pulses need their actual amplitude-level envelopes.

Work ledger and event labels

For $H(t) = H_0 + g(t)V$, unitary evolution implies

$$\frac{d}{dt} \langle H(t) \rangle = \left\langle \frac{\partial H}{\partial t} \right\rangle = \dot{g}(t) \langle V \rangle. \quad (19)$$

Matter is not energetically closed: each Born sideband has $\Delta E_{\text{matter}} = q\Omega = -\Delta E_{\text{pump}}$. With $W_{lmq} = w_l K_{lmq}$,

$$Z_D = \sum_{lmq} W_{lmq}, \quad \mathcal{K}_{\text{work}} = \sum_{lmq} W_{lmq} q\Omega, \quad \langle W_{\text{pump}} \rangle_{\text{scattering}} = \mathcal{K}_{\text{work}}/Z_D. \quad (20)$$

The work coefficient is not a power; luminosity/overlap is missing. No quantum pump apparatus, depletion or backreaction is derived. $m = n, q \neq 0$ preserves the mode but is *not elastic*; only $m = n, q = 0$ is elastic. A homogeneous drive supplies no compact momentum and does not rotate one free particle into another mass branch.

4 Readout with an unobserved supply channel

In the driven prediction use

$$h_{jq}(x) = w_{j+m-n} \int_0^\infty dp f_{\sigma_i}(p) K_{j+m-n,m,q}(p) \mathcal{R}_{\sigma_d}(x | p_{f,jq}), \quad h_j(x) = \sum_q h_{jq}(x). \quad (21)$$

The classical beam is diagonal in p , and different l give orthogonal recoil states. With the long-time sideband limit this justifies summing rates. It is not a treatment of finite coherent wavepacket interference. If a hypothetical independent readout also provided q , the optimal error would obey

$$e_*^{\text{tagged}} = \frac{1}{Z_D^{(m,r)}} \sum_q \int dx \min(h_{+q}, h_{-q}) \leq e_*^{\text{untagged}}. \quad (22)$$

This is an information bound, not a constructed pump measurement. The passive inversion is invalid without accounting for q . For known p, q , repair it by subtracting work:

$$l^2 = R^2[(E_f - q\Omega - E_n)^2 - M_D^2 - p^2], \quad j = \frac{l^2 - (m-n)^2 - r^2}{2(m-n)}. \quad (23)$$

At $p = .2$ the six momenta for the selected output are

j	$q = -1$	$q = 0$	$q = +1$
+1	.82821583	1.02175458	1.20257369
-1	1.18924247	1.36191317	1.52787997

The closest opposite-sign pair is separated by .01333122, versus the passive carrier gap .34015860. All six are still distinct: zero-noise catalogue identification at known p remains possible. At finite resolution use the full mixture, not just the smallest spacing. Exact crossing of these two sidebands occurs at

$$2\Omega_* = \sqrt{M_D^2 + 4/R^2 + p^2} - \sqrt{M_D^2 + p^2}, \quad \Omega_*(.2) = .19243951642. \quad (24)$$

This diagnostic is not used to tune the chosen drive. At the crossing, those opposite-sign contributions have identical momentum readouts unless additional information is measured.

5 Completion boundary and the next physical decision

The sharp-beam driven ledger contains 66 open (l, m, q) rows, split as (14, 25, 27) across $q = (-1, 0, +1)$. It gives $Z_D = 2.10563491267 \cdot 10^{-6}$, a mode-conversion fraction .87381701, and mean pump work .00900097215 per counted scattering event. These are not absolute encounter efficiencies or a power.

Selected optimal errors, in percent, illustrate the physical decision:

σ_i	σ_d	No drive	Drive, q unknown	Drive, ideal q tag
0	.1	3.55155	5.76199	3.55366
.1	.1	3.77384	5.94485	3.77589
.2	.1	4.84993	6.83823	4.85210
.4	.1	11.0294	12.2710	11.0331

Thus at $(.1, .1)$ the chosen 5% criterion passes without a drive, fails with unobserved pump exchange, and passes with an ideal tag. This does not make every driven setup worse, or supply a physical tag. At $\sigma_i = .1$, improving σ_d to .05 instead gives a driven untagged error of 2.20225%.

The reports retain all 50 resolution cards, conditional false-sign errors, selected priors and decision boundaries. Quadrature orders 24 and 48 and two boundary-search mesh widths are

compared. Gaussian CDF intervals, not sampled histogram areas, give the error integrals. For the retained incident region $z \leq 10$, the omitted probability is $\tau = \Phi_N(-10)/\Phi_N(p_0/\sigma_i)$. Since $p_f/E_{\text{out}} \leq 1/2$,

$$K_{lmq} \leq \frac{|g_q|^2}{32\pi m_n \mu_l}, \quad \delta Z \leq \tau \sum_{l \in \{0, -2\}, q} \frac{w_l |g_q|^2}{32\pi m_n \mu_l}. \quad (25)$$

An error-change bound is $2\delta Z/Z_{\text{retained}}$; the response-tail loss is at most $2\Phi_N(-10)$. Mesh agreement is a numerical convergence check, not a theorem that all possible mixture roots have been isolated for arbitrary cards. Gaussian tails are numerically bounded, not declared evidence that a five-dimensional EFT is valid at arbitrarily high energies; a UV validity scale and loop uncertainties remain unspecified. Results are tree-level, long-time conditional forward predictions, not a physical particle fit.

The next bounded decision is an actual readout specification: resolution, beam spread, mass confusion and any independent pump-energy information. Absolute four-dimensional collision probabilities are unnecessary for relative recoil discrimination. If finite pulses, coherent interference or absolute probabilities are desired, specify spatial envelopes, timing and relative pump phase together and compute their common amplitude. A momentum histogram alone does not supply those inputs.

References

- [1] Particle Data Group, *Kinematics*, Review of Particle Physics (2024), sections 49.3–49.5, <https://pdg.lbl.gov/2024/reviews/rpp2024-rev-kinematics.pdf>.
- [2] M. Moskalets and M. Büttiker, *Floquet scattering theory of quantum pumps*, Phys. Rev. B **66**, 205320 (2002), <https://arxiv.org/abs/cond-mat/0208356>.